Online Dashboard For Lab Assistants To better support students in labs

Abubakar Othman (F221611)
MSci Computer Science and Mathematics
Department of Computer Science
Supervisor: Dr Firat Batmaz
Module Code: 25COD290

January Deliverable 2026

Contents

1 Introduction

1.1 Problem

Modern Computer lab sessions often have many students who come in to work with others on tasks that may involve programming or simply answering questions, especially in this day and age, when more and more students are leaning towards quantitative subjects. During a computer lab session, a vast majority of students are working simultaneously, on different questions at different speeds, as some may pick up questions faster than others or come in with more prior knowledge. It is in our best interest as educators to provide timely, high-quality support to students, monitor their progress, and prioritise those who are struggling most so they do not fall behind. Achieving these goals is difficult in practice, which we aim to address in this paper.

In education, it is key to identify students who are struggling the most so that we can improve their learning outcomes and help them as soon as possible, preventing them from falling behind. Student struggle comprises confusion, impasse, and help-seeking behaviour. By leveraging mathematical modelling, we can gauge which students require the most assistance and attend to them promptly. Having a "student struggle" measure will be vital in computer labs, as it will help educators identify which students need the most assistance. Currently, there are no models to predict which students are struggling most in live sessions, emphasising the need to develop such a model.

At present, staff rely on surveying the computer lab to identify struggling students or on students asking for help on their own initiative. This approach has a range of limitations, such as:

1. Shy Students - some students may be hesitant to ask for help in a lab environment
2. Prioritisation - it is harder to identify students who are struggling with this approach
3. Lazy Lab Assistants - lab assistants are paid to help out, but may have no interest in actually helping and wish to do their job; because of this, they may walk around. By finding out what students are struggling with, we can make them actually help out by sending them to help

In practice, we can see that it is currently hard to detect patterns such as repeated failed attempts, long time spent on a question, which questions are actually the hardest and more. In larger modules, we can expect at least 40 students to turn up. Having identified such patterns, we will be able to work more efficiently, making it a win-win scenario for staff and students.

1.2 Proposed Solution

In this project, we will address these issues and limitations by proposing a real-time learning analytics dashboard that supports lab teachers and assistants during computer lab sessions. Rather than replacing teachers and assistants with AI feedback and teaching, we will leverage LLMs to complement human support by identifying struggling students and complex questions. We will also leverage mathematical and statistical modelling to identify struggling students and complex questions.

The system will take in live student data from an endpoint. Using this data, we will define a range of useful metrics to aid in providing the best possible dashboard. Metrics include the number of attempts per student, the time spent on questions, and the average number of attempts to complete a question. With these metrics, we will be able to prioritise students and questions efficiently.

Whilst the primary interface of the proposed system is a desktop-based dashboard intended for improving monitoring of computer labs, the design will allow us to extend the functionality to mobile phones so that we can send notifications to the phones of lab assistants to notify them of where they should go to help, this functionality will also have the capability of being able to be extended into smart watches and glasses. This will allow teachers and assistants to receive timely alerts without constantly checking the monitor for relevant information.

By enabling lab teachers and assistants to identify challenging questions and struggling students more quickly during computer lab sessions, we will allow more proactive, practical support. With this solution, we will achieve significantly better learning outcomes.

1.3 Aims and Objectives

The primary aim of this project is to design and implement a system that enables lab teachers and assistants to make well-informed, real-time decisions during computer lab sessions to support students and improve learning outcomes more efficiently.

The secondary aim is to extend our system to smart devices. Whilst the core system is a desktop-based dashboard, we can improve usability by sending notifications or messages to smart devices to enable easier communication with assistants.

The aims will be addressed through the following objectives:

1. **Identify relevant literature and existing systems** related to learning analytics, real-time data visualisation, and AI in education, to identify existing limitations in current approaches to relevant areas
2. **Formalise system requirements** by defining functional and non-functional

requirements for the real-time computer lab support system

3. **Design Data Model** that extracts relevant data provided into meaningful metrics suitable for real-time analysis
4. **Design system architecture** to support live data ingestion, prioritisation and presentation while remaining extensible to smart devices
5. **Develop a struggle and difficulty model** so that we can take the relevant values to produce metrics which will be used in showing the relevant information
6. **Implement live data analytics dashboard** for real-time data visualisation, prioritisation and intuitive interactivity
7. **Evaluate approach** through functional and non-functional aspects to determine the feasibility and usefulness in a computer lab session

Together, these objectives will guide the project toward becoming a full-fledged EdTech application that aims to improve learning outcomes by enabling more efficient and targeted support.

1.4 Risks and Mitigation

The project involves developing a real-time data analytics system for use during computer lab sessions. As such, potential technical and methodological risks have been identified and will be addressed through deliberate design decisions and extensive testing.

1.5 Project Approach

This project follows an agile approach that prioritises usefulness, practicality, and user focus for a computer lab setting. Given the reliance on the data endpoint and in-lab use, the strategy emphasises thorough data analysis and early testing.

The project begins with an analysis of relevant literature and existing systems to understand how learning analytics, real-time visualisation and AI-supported tools have been applied in education. The analysis will also allow for the identification of limitations in current approaches. The valuable insights gained from the study of this data will prove to be valuable in developing the functional and non-functional system requirements ensuring that the system to implemented in the best possible way.

Following this, we will then focus on the system design. This will include the development of the high-level system architecture, the design of the data model to extract and process relevant data and prioritisation models for student

Risk Description	Potential Impact	Mitigation Measures
Latency in real-time data processing	Delays in updating the dashboard can reduce the usefulness and impact of timeliness of support	Ensure there are lightweight analytics and incremental updates
Unstable struggle and difficulty estimates due to insufficient data	Irregular interactions may lead to incorrectly identifying struggling students	Establish a baseline for suitable data or use an LLM to identify whether data is suitable
Overly complicated dashboard	Overcomplicating the dashboard may be more distracting and be unintuitive	Seek constant feedback on the UI so that we can implement it correctly and focus on main metrics
Unable to implement smart glasses functionality	Due to technical constraints, we may not be able to fully integrate the dashboard to smart glasses	Explore smartwatch solutions or simplify what goes into the smart glasses
Limited targeted towards different modules	A model that works well on one model may not work well on another	Design models over generic interaction patterns rather than specific module indicators

Table 1: Identified Risks and Mitigation Strategies

struggle and question difficulty. These design activities will be conducted to account for noisy data and enable extensibility to implement connectivity to smart devices.

The implementation will be done incrementally, allowing us to focus on data connectivity and processing initially. Afterwards, we will design an initial dashboard to display the data visualisations, ensuring that the data-to-dashboard process is working correctly. Then we will implement core functionality, such as interactive graphs that display important information, including students who need help and complex questions. Thereafter, we will implement connectivity to smart devices, starting with smartphones, to notify lab assistants, then to smart watches and glasses.

Finally, we will evaluate our results by obtaining feedback on our dashboard design to ensure it is feasible for the targeted user, and by carrying out functional and non-functional testing to ensure everything is working correctly. The approach ensures that each phase of the project builds seamlessly on the previous one, resulting in clear project progression in line with the aims, objectives, and identified risks.

2 Background and Requirements

2.1 Literature

2.1.1 Learning Analytics in Higher Education

In 2012, the use of data analytics was shown to improve student success by providing real-time feedback. Learning analytics is the measurement, collection, analysis, and reporting of data about learners and their contexts, all to understand and optimise learning and the environment in which it occurs. The growth of online education since the 1990s has contributed significantly to the advancement of learning analytics, as student data can now be captured and analysed. Learning analytics allows educators to make data-driven decisions.

Learning analytics leverages data sources; many types of data sources have been explored and used to help educators. Relevant data sources include Massive Open Online Courses (MOOCs), clickstream data (records of user interactions, such as clicks and time spent), and course content. There are also data sources that, in the context of our project, we can get away without using, such as demographic data and gaze patterns. Whilst 35% of the studies used more than one data source, the majority of studies still used only one data source. It is argued that the use of multiple data types will contribute more effectively to learning outcomes.

The outputs of learning analytics have proven to be valuable for pedagogical decision-making. Use includes personalisation and feedback, which can tailor recommendations to students' behavioural profiles to promote diverse learning. Another use case is the VITAL project, which confirmed that flipped classroom models were effective, as students accessed materials before attending face-to-face sessions. However, several studies note challenges and limitations in learning analytics. These include how analytics leverage observable behaviours such as clicks and time online rather than cognitive processes. Learning analytics also raises ethical concerns, such as privacy, when moving solutions out of labs. Many restrictions must be determined, as protecting students' identities and data is a priority, which we shouldn't lose sight of when trying to improve learning outcomes.

With the current utilisation of learning analytics, there is yet to be an approach that leverages smart devices in computer labs, in particular, smart watches and smart glasses. By leveraging smart devices, we can significantly increase the efficiency of lab assistance. From a centralised point, we can allow the lab teacher to send instructions to lab assistants based on these analytics, such as which questions are hardest and which students need the most help, thereby improving learning outcomes.

2.1.2 Dashboards for Learning Analytics

Learning analytics have been leveraged to use dashboards as valuable tools for both educators and learners, such as making educators aware of subtle interactions in their courses and allowing learners to gain insight into their learning actions and the effects they may have. Because of capabilities such as these, dashboards are the primary representation of learning analytics, as they can help improve learning outcomes and make the most of the data generated by learning analytics.

SAM was an early dashboard developed on data from learning environments, with both teachers and learners as target user groups. SAM allowed teachers to separate students who were doing well from those at risk based on time spent and resource-use metrics. Teachers gain insight into the time learners spent on tasks and can compare with their estimates, whilst resource use provides insights on how and how often students used documents and tools, it also allows discovery of resources external to the course. Metrics could also be added, depending on data detail, to increase understanding of specific tasks. EMODA is a more recent approach to the learning dashboard; it takes into account emotion, which researchers have shown to play a key role in learning, regulation processes, and strong coupling to motivation and achievement. EMODA enables teachers to monitor learners' emotions during online learning sessions by analysing audio, video, self-reports, and interaction traces.

Edsight is a web-based learning analytics dashboard developed with responsive web design that accommodates different device types and is accessible to educators. When first logging in, teachers are presented with all the practical measure data they have collected thus far, the classrooms with the most data, and how many days have elapsed since they last logged data for analysis.

Figure 1: "Overview," "Report," and "Activity List" modes, displayed in desktop, tablet, and phone screens.

The report is the main dashboard in Edsight, where teachers can select filters such as date and classroom for display. Once a filter is applied, different results are displayed.

Figure 2: (L) A single-event report displayed in Edsight, with filters on the right side. (R) Multi-Day view, with the ability to compare patterns across classes and sections.

Most current approaches to the learning dashboard live in lab sessions, as there is more course-level monitoring than session-level monitoring. This also has many benefits, helping students on the spot and allowing them to get the most out of their computer labs.

2.1.3 Real-Time Data Analytics and Visualisation

Real-time analytics refers to the practice of collecting and analysing data as it is generated, with minimal latency between data collection and analysis. When we can process data as it comes in, we can gain insight and understand what is happening within a relevant context almost instantly. Such a practice is used in applications where data timeliness is critical, such as smart pricing and predictive maintenance. Traditional data analysis, where data is captured and stored, and insights are pushed out of storage. In contrast, real-time data analysis allows data to be analysed in real time. Real-time data analytics have proven to be highly valuable, as evidenced by the success of Netflix, which has perfected the art of predicting its customers' next favourite show.

Real-time analytics has emerged as a pivotal tool across multiple sectors, including retail, revolutionising decision-making processes and operational strategies. Retail giants such as Walmart and Amazon have leveraged real-time data analytics to personalise marketing campaigns, optimise pricing strategy, and efficiently manage inventory. Technologies such as RFID, IoT, and advanced analytics platforms were used to gather real-time data, analyse customer behaviour, and make data-driven decisions. A less utilised but more relevant application is in education, for example, a real-time machine learning application that detects learners' attention levels and participation using sensory data. The data allows teachers to recalibrate their teaching as necessary.

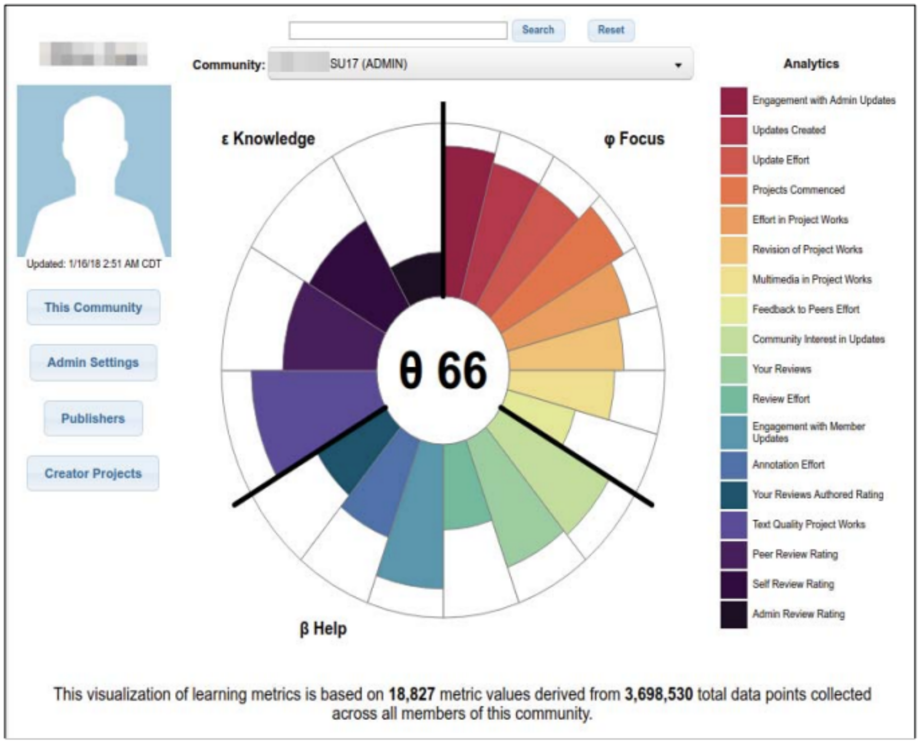

Figure 3: The Learning Analytics Interface for a dashboard which collects data based on student attention

Applying real-time analytics in computer labs will prove to be very beneficial for students and teachers alike. By collecting and processing data as it comes in, we will be able to update the dashboards in real time and provide teachers with the most up-to-date information as needed, enabling relevant, timely support for those who need it. This would be better than a traditional data analysis approach, in which we review the data after the lab has finished, because our findings may not be relevant by then.

2.1.4 Summary Identified Research Gaps

This subsection reviewed a range of relevant literature. Learning analytics has proved to be incredibly useful in higher education by providing data such as MOOC completion rates, click counts, and time spent, which have significantly aided pedagogical decision-making. Learning dashboards have also emerged as the primary output of learning analytics, enabling educators to gain insight into how different students are performing through an intuitive user interface. Real-time analytics has enabled processing data as it arrives, benefiting multiple sectors. Although all this work has proven highly beneficial in education and other sectors where data use is extremely valuable, there are various gaps we will address in this thesis.

In particular, computer labs have a distinct challenge that isn't well addressed by existing approaches. In these settings, teachers and assistants have no way to swiftly identify those who need help and must rely on those who put their hands up. The problem with this is that it does not account for shy students who may not want to put their hands up, causing them to spend too much time on a question. Also, current dashboards focus on data collected towards the end of the lab; this data may not be as valid afterwards. By creating a dashboard that uses real-time analytics, we can efficiently leverage this data to support students during lessons.

Furthermore, there has been no use of AI to classify data and provide a new metric, which could remove the need for educators to read every solution. This metric can be used to assess how close the student is to the correct answer and can provide a valuable indicator of how much a student is struggling. Combining this with a range of other metrics, such as attempts and time spent, will help improve student outcomes. There is also a lack of dashboards that integrate with smart devices in computer labs. This can be used to send data from the dashboard to lab assistants' smart devices, such as phones, watches, or even glasses, to indicate which students they can help. This motivates the development of a real-time learning analytics dashboard that supports instructor awareness and prioritisation through smart device integration.

2.2 Existing Systems

2.2.1 Learning Management System

A learning management system is a software application used for the administration, documentation, tracking, reporting, automation, and delivery of educational courses, training programs, materials or learning and development programs. Learning management systems also constitute the largest share of the learning systems market; almost all higher education institutions have adopted them. They were designed to identify training and learning gaps using analytical data and reporting. A notable learning management systems is Moodle which is the world's most popular learning management system.

A limitation of these systems are that the analytics are typically reviewed after activities, meaning some data may not be as useful after and students that were struggling in computer labs may have not be provided timely assistance. Another limitation of such systems is that they are primarily designed for course level and while this is useful for curriculum design and long-time planning, they are generally not optimised for in lab computer sessions where timely support is critical.

2.2.2 Instructor Facing Dashboards

Learning management systems often incorporate dashboards to track student or user progress, helping teachers better understand knowledge gaps. Dashboards provide insight into student progress through visualisation and scores derived from data generated during teaching and learning activities. As a result, dashboards are at the centre of learning analytics, integrated with learning management systems.

Existing learning dashboards, such as Blocks: Progress Bar, report on student progress rather than student struggle at the lab level. Dashboards like these show user active time, assignment progress, learning progress, marks, and assignment submissions. Visual components such as the progress bar aid in the understanding of student progress, removing all the complexity that is going on behind the scenes. Although these are useful, they don't directly present the user with more complex questions compared to others, and students struggle. Dashboards provide

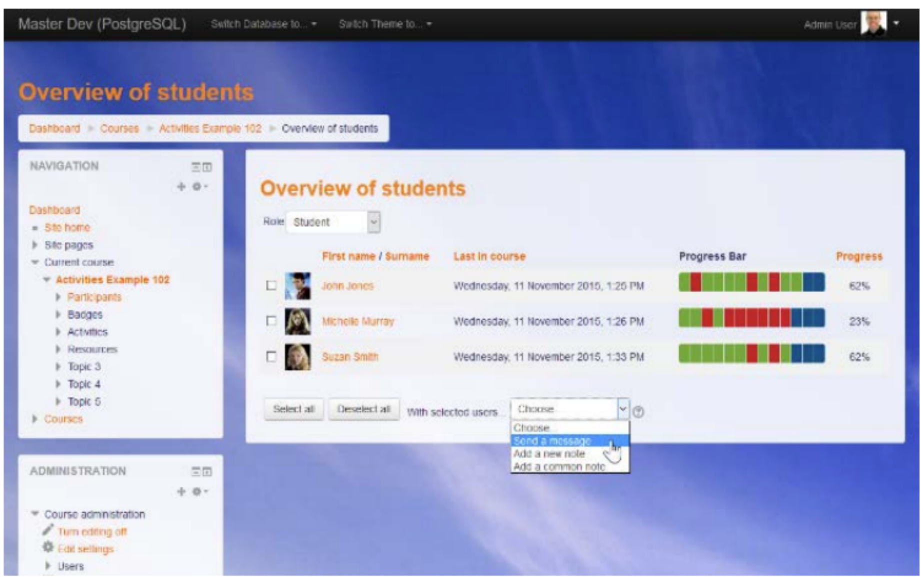

Figure 4: Piwik Analytics

Another implementation of a learning dashboard is the Multi-Modal (MM) teacher dashboard, the MM dashboard: 1) receives data from eye trackers, wristbands, web cameras, and the learning activity; 2) computes the student's relevant cognitive, affective, and physiological measurements; 3) synchronises and visualises the resulting measurements in a clean, customisable interface; and 4) notifies instructors during potential moments of interest.

Figure 5: Different views of MM dashboard: SSV (left), ASV (centre), and RSV (right).

The MM dashboard has three ways of data visualisation, as we can see in 5: 1) single session view (SSV), which displays real-time learning data from a single student's session; 2) all sessions view (ASV), which shows real-time learning data from all connected students; and 3) replay session view (RSV), which allows the teacher to interact with previously recorded session data from a single student session. The views are divided into panels, and we can also see that the SSV and ASV have a right-side navbar used to switch between panels.

However, the dashboards do come with a range of issues. For the MM dashboard to work, students must have a range of sensors, which are uncommon for them. Teachers also report no experience with some of the hardware required for the MM dashboard to work. Although this is a key problem, it was not the MM dashboard's main issue; the main issue was that teachers may lack the knowledge to understand and interpret the dashboard results, as the majority of users reported unfamiliarity with the measurements provided. The importance of the metrics was also questioned. The study also noted that hands-on training may be required to help teachers use the system efficiently.

The limitations highlight a critical gap in the practicality and potential for utilisation of the learning dashboards. Whilst the dashboards provide high-level insights, many of these insights are difficult to understand, prompting educators to question their importance. Also, these dashboards tend to be unintuitive, which can be especially an issue for educators who are not STEM-oriented. We can leverage these insights to create clearer indicators, such as struggle, and by taking a more straightforward approach to the dashboard, we can address these gaps and create better, more useful learning dashboards.

2.2.3 Early Warning and At-Risk Student Systems

Early warning systems are a chain of information communication systems that can be implemented, comprising sensors, event detection and decision subsystems for early identification. An At-risk student system is an early warning system that spots struggling students before it is too late. Recently, many approaches to at-risk student systems have leveraged AI to identify students at risk.

Student data is analysed to flag potential issues. Typically, grades, attendance, online activity, and behaviour are used to alert teachers to help students early, before problems grow. The datafication of education enables the use of automated methods to detect patterns in extensive educational data. Approaches leverage predictive models to use the data, predicting the likelihood that a student will fail a course.

We can look at the edInsight Early Warning System as an example. This system looks at the course as a whole and uses the information gained to determine an at-risk ranking for all students. The system leverages a scoring system, based on specific categories such as the Course grade category:

Grade	Score
C-	2
D+	4
D	8
F	10

Table 2: Scoring system based on grades, where lower grades add to the score, showing that a student is struggling more

With the scores calculated, educators have a clear idea of which students are most at risk, making this information useful for directing attention to those who need the most help.

Although the system is beneficial for identifying students who are struggling at the course level, it does not necessarily help identify students at the lab level, as its design assumptions do not align with the demands of lab sessions. Educators working in real-time sessions require live data. Hence, they have immediate awareness of which students are currently struggling, rather than future failure. However, we can apply the same concepts used in at-risk systems to identify students who are struggling the most and questions that have the most difficulty. As a result, real-time systems may not be suitable for labs, but they provide a key framework for the approach we can take to help students in computer labs.

2.3 Requirements Specification

2.3.1 Functional Requirements

In this section, we define the core functionality required of the proposed system to address the problem in Chapter 1 and the gaps highlighted earlier. These requirements specify what the system must do regardless of implementation.

FR1: Live ingestion of Lab data The system must be capable of ingesting student interaction data as it is generated in the lab sessions, which includes data which is produced whilst students are answering questions, such as attempts amount, how wrong the answer is, how long spent on the question, which will be used for measuring student struggle as well as metrics we will use in measuring question difficulty such as total attempt amount, total correct submissions and average time spent.

FR2: Computation of student struggle The system must compute a metric to represent how much a student is struggling in labs, derived from data highlighted in **FR1**. This representation should represent short-term difficulty rather than long-term ability and predicted outcomes.

FR3: Computation of question difficulty The system must compute a metric for question difficulty used to represent how hard a question or task is in a lab session, derived from the data highlighted in **FR1**. This will represent a task that is difficult for everyone in attendance at the lab.

FR4: Real-time prioritisation The system must provide an indicator of which students are struggling most in computer labs, based on student struggle data, allowing instructors to aid those who need it most.

FR5: Present Relevant Analytics The system must present relevant analytics for the labs in a suitable, intuitive form that can be interpreted quickly to enable quick decision-making.

FR6: Smart Device The system must integrate with smart devices such as phones, watches and glasses, in particular the smartphones so that information can be sent swiftly to lab assistants.

FR7: Lab Assistant Ranking System The system will comprise a ranking system of lab assistants to incentivise them to help students. The ranking will be based on the number of students they have helped and the students' satisfaction with their help.

2.3.2 Non-Functional Requirements

In addition to functional requirements, the system must satisfy non-functional requirements to ensure the system operates as intended. These requirements reflect the practical constraints of the live lab sessions and ensure efficient operation.

NFR1: Real Time Performance The system must process and update analytics with minimal latency to support swift decision-making in lab sessions. Delays should not prevent an instructor from responding to students who need help.

NFR2: Interpretability Outputs should be interpretable by instructors without requiring specialist knowledge or extensive training. The system should be intuitive so that it can be used across all educational disciplines. Analytics should aid in understanding rather than providing overly complex information.

NFR3: Robustness to noisy data The presence of noisy or incomplete data should not interfere with the system's operations. Noisy or incomplete data should be accounted for and not interfere with the provision of correct analytics.

NFR4: Scalability The system should be capable of operating efficiently in various labs regardless of module, class size and question type.

NFR5: Privacy and Ethical Data must be handled responsibly and solely for the purpose of providing analytics in computer labs, and it must not be used for any purpose other than aiding students.

NFR6: Extensibility and Maintainability The system must be designed to accommodate additional data sources or metrics without requiring a fundamental redesign, enabling future extensions.

2.3.3 Requirements Prioritisation (MoSCoW)

For this section, we will use a MosCoW table to highlight the project's requirements prioritisation, reflecting on time constraints and the project scope.

Must Have	Should Have	Could Have	Won't Have
FR1	FR5	FR7	Long term student outcomes
FR2	FR6		Fully AI decision making
FR3	NFR3		Avatar based AI assistance
FR4	NFR4		
NFR1	NFR6		
NFR2			
NFR5			

Table 3: MoSCoW table highlighting requirements for proposed software

3 Design and Modelling

This section describes the design and architecture of the lab dashboard, highlighting how various components will be implemented. Additionally, we will talk about the data endpoint, defining what data enters the system and in what form. We will then discuss the mathematical models we plan to leverage for the project. Finally, we will discuss how we will expose the analytics to educators to provide an intuitive solution.

3.1 System Architecture

The proposed system adopts a layered architecture designed to support real-time learning analytics in computer lab environments. The architecture separates data generation, data ingestion and processing, as well as decision-making and action, into three distinct layers, ensuring each component can evolve independently. Although this focuses on an interval-based approach to the data, an event-driven pipeline is currently under exploration.

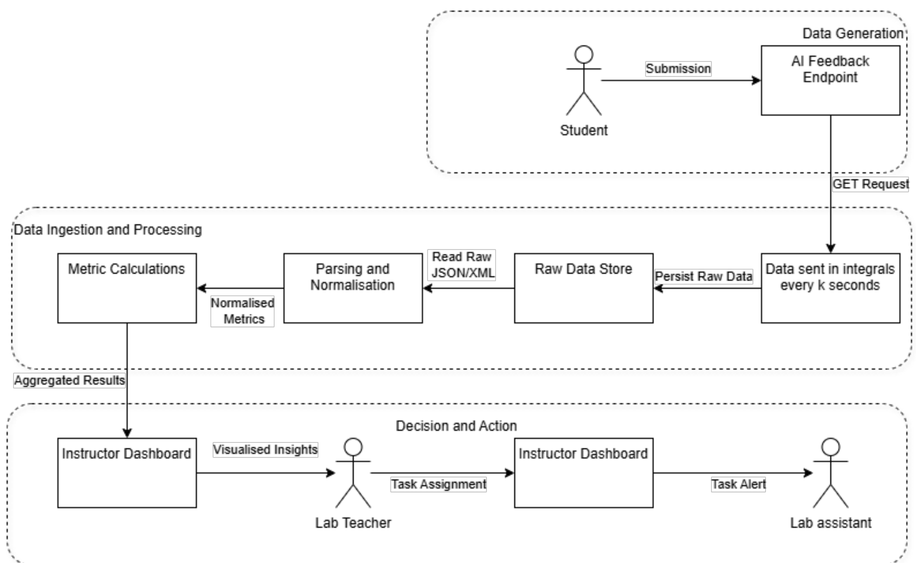

Figure 6: Diagram showing the architecture of the proposed system

Data Generation Layer As seen in Figure 6, student interactions form the primary source of data. When students submit an attempt or request feedback, the interaction data is recorded at the endpoint and not modified.

Data Ingestion and Processing Layer Data ingestion is performed periodically according to the user-defined refresh rate. Retrieved data will be stored in the raw data store, then parsed and normalised before being used for calculations. The data will be used to determine student struggle, question difficulty, and other required metrics.

Decision and Action Layer Data Analytics will now be sent to the learning dashboard for presentation to the instructor, thereby aiding decision-making. The instructor will identify students who are struggling. More complex questions will also be identified so that the instructor can go through the questions with the whole class. Instructors will use the dashboard to assign tasks to assistants to assist students who need help.

This layered design supports the project's aim of providing timely, targeted support during lab sessions while using the system as a decision-making tool for student support. Later sections build on this architecture.

3.2 Data Endpoint

A data endpoint is a specific URL or address in a web service or API that defines where and how clients can access the service. The proposed system integrates with the existing AI feedback platform through a single external data endpoint. The data endpoint is crucial in our system as it provides access to detailed records of student submissions and feedback interactions generated as students engage with lab questions. As outlined in Figure 6, the endpoint serves as the entry point to the analytics pipeline, supplying the raw data that will be processed, normalised, and aggregated for data visualisation. The integration is designed to be read-only, ensuring the data and metrics are not tampered with.

The endpoint returns an array in JSON format, where each record corresponds to a specific session based on the session's first submission time. Alongside standard fields such as module, question, timestamp and user, each record has an XML payload. The payload captures the interaction history for a given session, including submissions and AI responses. The design reflects the constraints of the existing system while providing vital information for our analytics.

Interaction data are organised around sessions, where a session represents a continuous period of student engagement from the first submission attempt. Within a session, students may have multiple submissions, getting multiple AI responses. This is beneficial in reflecting how their answers may improve question by question. The structure allows the capture of temporal patterns, repeated attempts, and help-seeking behaviour, which will be vital for identifying short-term struggle and difficult questions in lab sessions.

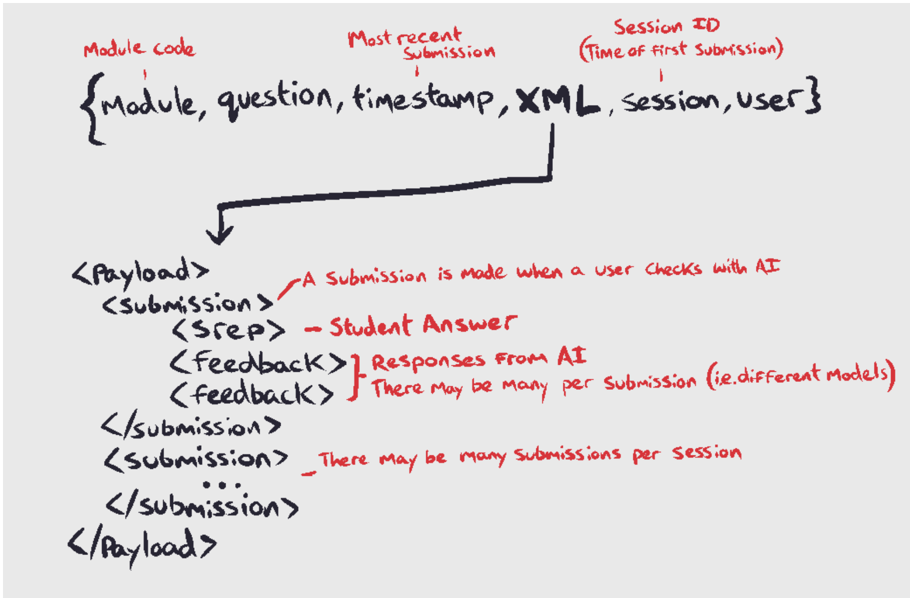

Figure 7: Structure of session-based interaction data returned by feedback system

From the session-based data, the system derives a set of core metrics for real-time monitoring. These include attempts per question, time between attempts, AI-based indicators of correctness, and the frequency with which students request feedback. By using these, we will enable the system to distinguish among students and identify those who are struggling the most. The derived metrics will form the basis of our two key metrics: student struggle and question difficulty.

It is essential to consider that we may receive incomplete or noisy data during sessions, which could stem from students repeatedly returning to the AI feedback system or from procrastination on their work. Not all indicators of struggle and question difficulty are valuable on their own; hence, they are combined to produce indicators of student struggle and question difficulty.

3.3 Mathematical Modelling

In this section, we will describe what we will measure and what available data will aid in measuring these.

3.3.1 Modelling Student Struggle

Modelling Objective To model student struggle, let s denote a student and σ denote the session. We will denote students who struggle with a scalar function:

$$S(s, \sigma) \in [0, 1] \tag{1}$$

The student struggle function will allow us to quantify the degree of difficulty that students s are currently experiencing during session σ, based on the observable data from the endpoint in Section 3.2. The model will support real-time prioritisation in computer labs.

Identified Variables From the session-based interaction data described in Section 3.2. We can derive the following quantities:

- $n_{s,\sigma}$ - The total number of submissions made by a student s during session σ
- $t_{s,\sigma}$ - The total active time elapsed during session σ
- $e_{s,\sigma}$ - The number of incorrect submissions with session σ
- $f_{s,\sigma}$ - The number of AI feedback requests within the session σ

By leveraging metrics linked to student struggle, we can produce a student struggle score

AI-Based Answer Rating Let $A_{s,\sigma,k}$ be the k-th submitted answer by student s in session σ. We will derive an LLM-based scoring function. Where Q_σ is the current question and A_σ is the answer, we will define our LLM-based scoring function to be:

$$c_{s,\sigma,k} = C(A_{s,\sigma,k}, A_\sigma, Q_\sigma) \in [0, 1] \tag{2}$$

where $c_{s,\sigma,k}$ is the correctness measure that measures the quality of a students answer, where 1 is the correct answer and 0 is completely wrong. We can then define our incorrectness measure:

$$i_{s,\sigma,k} = 1 - c_{s,\sigma,k} \in [0, 1] \tag{3}$$

Normalisation We will define maximum and minimum amounts as such:

$$n_{min} = \min_{s \in S} n_{s,\sigma}, \quad n_{max} = \max_{s \in S} n_{s,\sigma}$$

We can apply the same definitions to other metrics where relevant. To ensure we can compare the metrics, it is necessary to normalise our quantities:

$$\hat{n_{s,q}} = \frac{n_{s,\sigma} - n_{min}}{n_{max} - n_{min}}, \quad \hat{t_{s,q}} = \frac{t_{s,\sigma} - t_{min}}{t_{max} - t_{min}}$$
$$\hat{e_{s,q}} = \frac{e_{s,\sigma}}{n_{s,\sigma}}, \quad \hat{f_{s,q}} = \frac{f_{s,\sigma}}{n_{s,\sigma}}$$

Hence,

$$\hat{n_{s,\sigma}}, \hat{t_{s,\sigma}}, \hat{e_{s,\sigma}}, \hat{f_{s,\sigma}} \in [0,1]$$

Give recent submissions more importance To reflect that recent submissions are the most relevant thing in the live setting, we will aggregate the most recent m submissions using convex weights w_i:

$$A^{raw}(s,\sigma) = \sum_{i=0}^{m-1} w_i i_{s,\sigma,k-i}, \quad w_i \geqslant 0, \sum_{i=0}^{m-1} w_i = 1 \tag{4}$$

Since $i_{s,\sigma,k-i} \in [0,1]$, it follows that:

$$A^{raw}(s,\sigma) \in [0,1]$$

Struggle Score Definition Define the raw struggle score as a convex combination of the defined bounded components:

$$\mathbf{S}^{\text{raw}}(s,\sigma) = \alpha\hat{n}_{s,\sigma} + \beta\hat{t}_{s,\sigma} + \gamma\hat{e}_{s,\sigma} + \delta\hat{f}_{s,\sigma} + \eta A^{\text{raw}}(s,\sigma), \tag{5}$$

where:

$$\alpha, \beta, \gamma, \delta, \eta \geqslant 0, \quad \alpha + \beta + \gamma + \delta + \eta = 1$$

Since we are using normalised values it follows that:

$$S^{raw}(s,\sigma) \in [0,1]$$

Temporal updating for real-time operation To provide a responsive, stable indicator, we will update struggle with exponential smoothing:

$$S_t(s,\sigma) = (1-\lambda)S_{t-1}(s,\sigma) + \lambda S^{raw}(s,\sigma), \quad \lambda \in (0,1] \tag{6}$$

It follows that the update preserves boundaries, giving:

$$S_t(s,\sigma) \in [0,1], \quad \forall t$$

Interpretation of model Higher values of $S_t(s,\sigma)$ indicate that student s is experiencing sustained difficulty in the current session. The metric in Equation 3.3.1 will prove to be valuable in real-time prioritisation of help in lab sessions. This measure does not indicate student ability in the cohort but rather in lab struggle.

3.3.2 Modelling Task Difficulty

Modelling Objective To model question difficulty, let q denote a question during the lab session, and let $\mathcal{S}_q$ denote the set of student-session pairs associates with task q. We will denote questions that are difficult with a scalar function:

$$D(q) \in [0, 1] \tag{7}$$

The question difficulty function will allow us to quantify the degree of difficulty of a question q, that the students in the $\mathcal{S}_q$ set are finding the hardest, based on the observable data from the endpoint in Section 3.2. The model will support real-time prioritisation in computer labs

Identified Variables From the session based interaction data described in Section 3.2. We can derive the following quantities:

- n_q - The total number of attempts on question q
- c_q - The total number of correct attempts on question q
- t_q - The average time spent on question q
- a_q - The average number of attempts spent on question q
- f_q - Average AI incorrectness measure for q, derived from AI-Based answer ratings

By leveraging metrics linked to question difficulty, we can produce a question difficulty score.

Normalisation To ensure we can compare the metrics, it is necessary to normalise our quantities:

$$\tilde{c_q} = 1 - \frac{c_q}{n_q}$$
$$\tilde{t_q} = \frac{t_q - t_{min}}{t_{max} - t_{min}} \quad \tilde{a_q} = \frac{a_q - a_{min}}{a_{max} - a_{min}}$$

f_q is already bounded so:

$$\tilde{f}_q = f_q \in [0, 1]$$

Difficult Task Definition Define question difficulty as a convex combination of the bounded components:

$$D^{\text{raw}}(q) = \alpha\tilde{c}_q + \beta\tilde{t}_q + \gamma\tilde{a}_q + \delta\tilde{f}_q \tag{8}$$

where:

$$\alpha, \beta, \gamma, \delta \geqslant 0, \quad \alpha + \beta + \gamma + \delta = 1$$

Since we are using normalised values it follows that:

$$D^{raw}(q) \in [0, 1] \tag{9}$$

Temporal updating for real-time operation To provide a responsive stable indicator, we will update question difficulty with exponential smoothing:

$$D_t(q) = (1 - \lambda)D_{t-1}(q) + \lambda D^{raw}(q), \quad \lambda \in (0, 1) \tag{10}$$

Interpretation of Model Higher values of $D_t(q)$ indicate that a question is difficult for the class as a whole during the current session. The metric will prove to be valuable in real-time prioritisation of help in lab sessions.

3.4 Interaction and UI Designs

This section describes how the output of our models defined in Section 3.3.1 and 3.3.2 is presented to instructors in a way that enables quick decision-making during lab sessions. The design is intended to provide an intuitive view.

3.4.1 Design Principles

The user interface is driven by the requirement to expose the model outputs defined in Sections 3.3.1 and 3.3.2 in a form that enables rapid, intuitive interpretation and decision-making. Unlike the previous dashboard we explored, this one will be more intuitive and will be used primarily for computer labs.

The primary objective is to present student struggle and question difficulty as distinct signals. Student struggle indicators are derived from the struggle score $S_t(s, \sigma)$ and question difficulty indicators are derived from the difficulty score $D_t(q)$. These two indicators will prove to be valuable in helping educators improve learning outcomes.

A further objective is to allocate lab assistants to students so they can help promptly. This will send a notification to the lab assistant allocating them to students. Another is avoiding exposure of raw data, which has been mitigated by data normalisation.

Finally, the system focuses on supporting decision-making rather than explanatory analysis. The dashboard is intended to guide instructors towards timely intervention by highlighting students and tasks requiring attention and enabling efficient allocation of lab assistants.

3.4.2 Instructor Facing Dashboard

The dashboard provides a real-time overview of student struggles and question difficulty during the computer labs. The dashboard is will allow the instructor to make quick decisions and aid those in need. To illustrate the proposed interface, a dashboard mock-up was created using Figma. The mock-up represents a conceptual design rather than a fully implemented system.

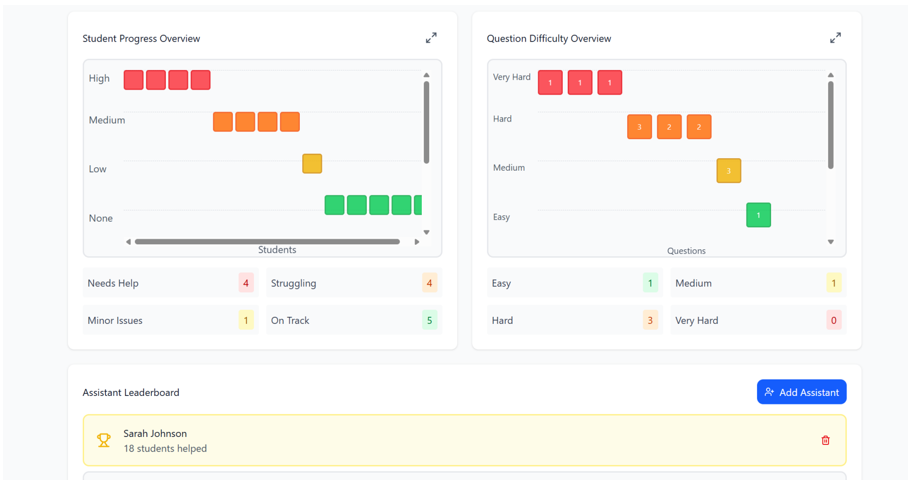

Figure 8: Views based off student struggle and question difficulty

As shown in Figure 8, our dashboard has two key views: the first shows a list of students whose struggles are at certain thresholds, hence the colour; the second shows a list of questions based on difficulty at a threshold, thus the colour. The values used for comparison against the thresholds are derived from our student struggle and question difficulty equations. The view also allows instructors to go over additional information.

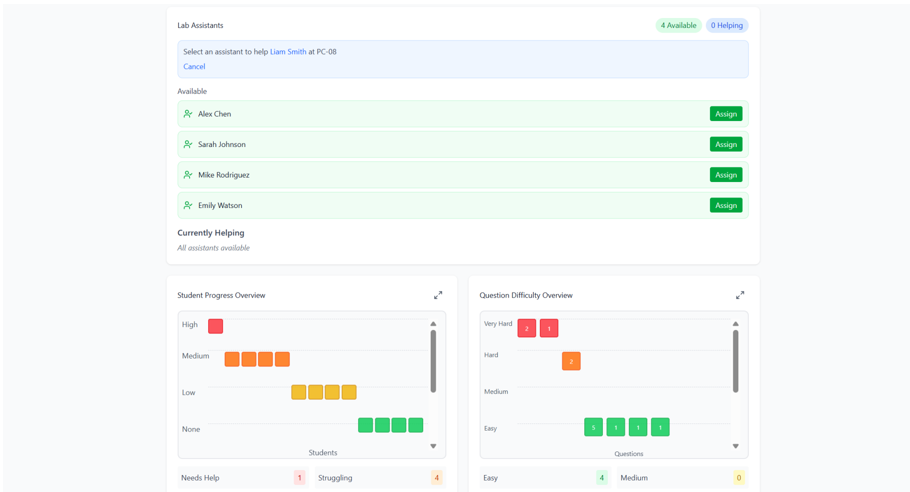

Figure 9: View of allocation of Lab Assistant

In Figure 9, we can select a lab assistant and then press a student to allocate them. This is a simple approach and a very minimal, intuitive design that allows use without training on the system.

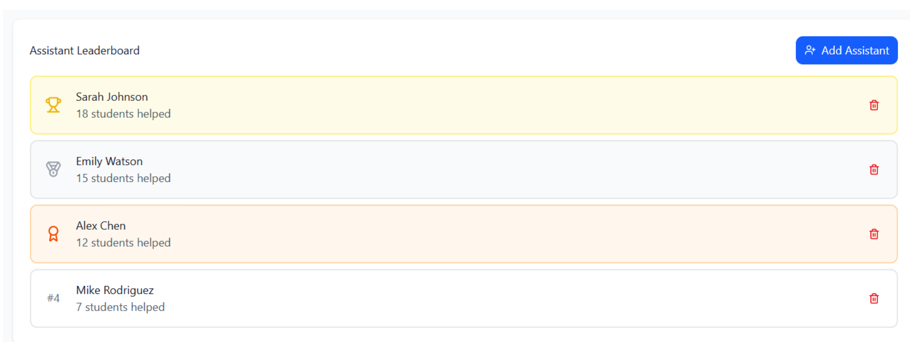

Figure 10: Assistant leaderboard, based on how many students have been help

In Figure 10, we can see that lab assistants are displayed based on their contributions. By gamifying the helping process, we can incentivise assistants to help more students.

3.4.3 Visual Encoding of Metrics

By deciding on thresholds, we can highlight a student by degree of struggle or highlight a question by degree of difficulty

Degree	None	Low	Medium	High
Threshold	$0 \leqslant S_t(s, \sigma) \leqslant t_{s1}$	$t_{s1} < S_t(s, \sigma) \leqslant t_{s2}$	$t_{s2} < S_t(s, \sigma) \leqslant t_{s3}$	$t_{s3} < S_t(s, \sigma) \leqslant 1$
Colour	Green	Yellow	Orange	Red

Table 4: Table Highlighting how thresholds can be indicated for struggle

Degree	Easy	Medium	Hard	Very Hard
Threshold	$0 \leqslant D_t(q) \leqslant t_{s1}$	$t_{s1} < D_t(q) \leqslant t_{s2}$	$t_{s2} < D_t(q) \leqslant t_{s3}$	$t_{s3} < D_t(q) \leqslant 1$
Colour	Green	Yellow	Orange	Red

Table 5: Table Highlighting how thresholds can be indicated for difficulty

4 Implementation

This chapter describes the first implementation (Version 1) of the proposed real-time lab support system. The purpose of Version 1 was to gain an understanding of the relevant tools, data pipeline and validate practical integration. At this stage, the implementation focuses on fundamental indicators rather than the full analytical models. Version 1 therefore serves as the foundation for what the dashboard will eventually become.

4.1 Scope of Implementation

The implementation represents Version 1 of the dashboard. Version 1 was implemented to ensure end-to-end functionality worked as planned and to experiment with data visualisation rather than to deploy the whole system proposed.

Version 1 focused on establishing a working data pipeline capable of processing live interaction data during labs, aggregating student activity at the lab level, and presenting indicators in near real time. This provides the foundation for the proposed system. By working on Version 1, I have familiarised myself with the relevant technology and the data pipeline.

4.2 Technology Stack and Software Selection

System Component	Technology Used	Purpose and Justification
Data Source	PHP-based HTTP endpoint	Existing lab infrastructure exposes student interaction as an endpoint
Data Format	JSON with embedded XML	JSON allows lightweight transport, whilst XML stores structured session-level submission
Data Processing	Python	Provides strong support for parsing of JSON data and performing numerical computations
Analytics Logic	Python	Version 1 has simple indicators for student struggle based on thresholds
Dashboard Framework	Streamlit	Enables fast development of interactive dashboards
Visualisation library	Plotly	Provides interactive visualisations which are intuitive
Deployment Environment	Personal Computer	Suitable for testing and prototyping

Table 6: What was used for each technical aspect and how

The stack highlighted in Table **??** enables a unified development workflow, in which data ingestion, processing and visualisation are implemented within a single Python environment. Whilst Version 1 relies on refreshing, the architecture will be largely consistent with the proposed system.

4.3 Data Pipeline Implementation

Version 1 implements an interval-based data pipeline designed to support near real-time monitoring during lab sessions. The pipeline proceeds as follows:

1. **Data Retrieval:** At fixed intervals the Python application re-requests the full dataset from the existing PHP endpoint. This refresh is implemented within the Python code. The refresh helps ensure that metrics are up to date.
2. **Data Parsing:** The retrieved response, formatted as JSON with embedded XML strings, is parsed within Python. Relevant fields such as question, student and timestamps are extracted
3. **Data Structuring:** Parsed records are transformed into an immediate representation suitable for analysis
4. **Metric Computation:** A naive metric approach was used, where thresholds are used for the attempt amount of a question. This was done to demonstrate the concept
5. **Dashboard Update:** The dashboard is refreshed using the newly computed values, providing an updated view of lab activity each time the Python refresh executes

More advanced modelling described in the Design Chapter is not yet integrated and is reserved for the proposed system.

4.4 Dashboard Implementation

The Version 1 dashboard provides a live visual overview of student activity. It is implemented as a Streamlit application that integrates data retrieval, metric computation and visualisation in a Python workflow.

The primary aim of the dashboard is to support fast awareness for instructors and lab assistants. Visualisations focus on simple, predictable indicators, some of which are not relevant but are a proof of concept and testing of the data. The indicators are presented through interactive Plotly charts, providing an intuitive user interface.

Basic interactivity is supported through Streamlit controls, enabling users to switch between different views. The dashboard refreshes automatically, following each refresh cycle, ensuring up-to-date information is displayed.

At this stage, the dashboard serves as a functional prototype for testing and validating the feasibility of live monitoring during labs. Most advanced features, such as model-driven prioritisation, assistant allocation and smart device notifications, are to be implemented in the later iteration.

5 Progress Report

5.1 Progress Status Overview

For the January milestone, the project is progressing smoothly. The Literature Review and Design Sections have been completed. A working Version 1 prototype has been implemented. Student Struggle and Question Difficulty models have been developed

5.2 Work Completed to Date

The following work has been completed as part of the project:

- A relevant literature review covering learning analytics, real-time dashboard and existing education systems
- Formulation of functional and non-functional system requirements
- Design of overall system architecture, including the data flow and component interactions
- Definitions of mathematical models for student struggle and question difficulty
- Implementation of Version 1 data pipeline using refreshing and session-level aggregation
- Development of a working instructor-facing dashboard prototype using Streamlit

This work establishes a solid foundation for future work.

5.3 Current Limitations

The version 1 dashboard comes with a range of limitations:

- Struggle is currently based on a single metric as a proof of concept
- AI Based classifications and question difficulty have not been implemented
- Data ingestion relies on refreshing rather than an event-driven architecture
- No task allocaiton to lab assistants

5.4 Planned work

The remaining work will focus on:

- integrating the proposed analytics into the existing pipeline
- replacing refresh ingestion logic with an event-driven logic.
- survey from both students and teachers
- Task allocation to lab assistants through smart devices

5.5 Gantt Chart

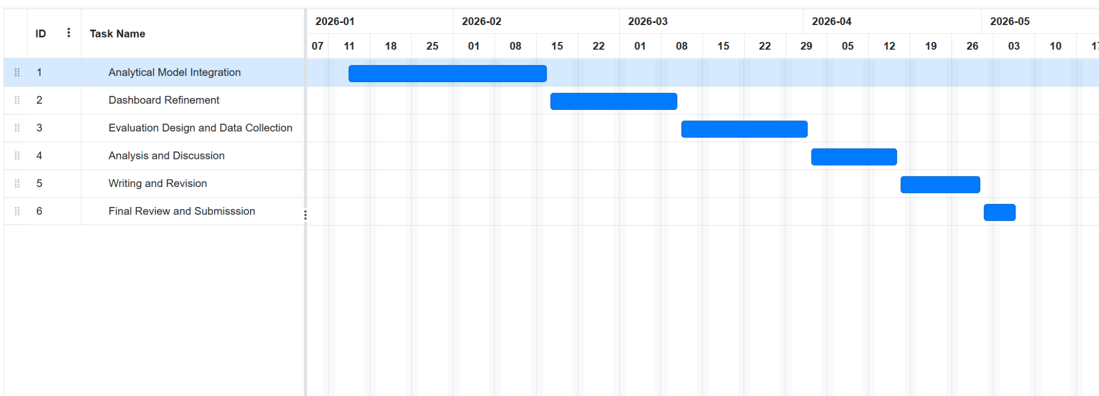

Figure 11: Project timeline from January 2026 to final submission in May 2026

Task	Start Date	End Date
Analytical Model Integration	14/1/26	16/2/26
Dashboard Refinement	17/2/26	10/3/26
Evaluation Design and Data Collection	11/3/26	1/4/26
Analysis and Discussion	2/4/26	16/4/26
Writing and Revision	17/4/26	30/4/26
Final Review and Submission	1/5/26	6/5/26

Table 7: Summary table for what is in the Gantt Chart

A Themes and References

Theme	References
Learning Analytics in Higher Education	[1–6]
Dashboards for Learning Analytics	[7–10]
Real-Time Data Analytics and Visualisation	[11–14]
Learning Management System	[1]
Instructor Facing Dashboards	[15, 16]
Early Warning and At Risk Student Systems	[17, 18]

Table 8: Table highlighted the themes covered and the references used in them

References

[1] W. Contributors, "Learning analytics," 05 2019.

[2] A. Nguyen, L. Gardner, and D. Sheridan, "Data analytics in higher education: An integrated view," *Journal of Information Systems Education*, vol. 31, pp. 61–71, 2020.

[3] C. Vieira, P. Parsons, and V. Byrd, "Visual learning analytics of educational data: A systematic literature review and research agenda," *Computers amp; Education*, vol. 122, pp. 119–135, 07 2018.

[4] A. Wilson, C. Watson, T. L. Thompson, V. Drew, and S. Doyle, "Learning analytics: challenges and limitations," *Teaching in Higher Education*, vol. 22, pp. 991–1007, 05 2017.

[5] A. Gelan, G. Fastré, M. Verjans, N. Martin, G. Janssenswillen, M. Creemers, J. Lieben, B. Depaire, and M. Thomas, "Affordances and limitations of learning analytics for computer-assisted language learning: a case study of the vital project," *Computer Assisted Language Learning*, vol. 31, pp. 294–319, 01 2018.

[6] D. Tzimas and S. Demetriadis, "Ethical issues in learning analytics: a review of the field," *Educational Technology Research and Development*, vol. 69, pp. 1101–1133, 03 2021.

[7] J. Klerkx, K. Verbert, and E. Duval, "Learning analytics dashboards," *Handbook of Learning Analytics*, pp. 143–150, 05 2017.

[8] K. Verbert, X. Ochoa, R. De Croon, R. A. Dourado, and T. De Laet, "Learning analytics dashboards," *Proceedings of the Tenth International Conference on Learning Analytics Knowledge*, 03 2020.

[9] S. Govaerts, K. Leuven, E. Duval, and K. Verbert, "The student activity meter for awareness and self-reflection."

[10] J. Ahn, F. Campos, M. Hays, and D. Digiacomo, "The journal of learning analytics works under a creative commons license," vol. 6, pp. 70–85, 2019.

[11] "Real time analytics," 06 2023.

[12] B. Marr, "What is real-time data analytics (and why it's so important)?," 07 2021.

[13] M. A. Raji, H. B. Olodo, T. T. Oke, W. A. Addy, O. C. Ofodile, and A. T. Oyewole, "Real-time data analytics in retail: A review of usa and global practices," *GSC Advanced Research and Reviews*, vol. 18, pp. 059–065, 03 2024.

[14] M. Montebello, "Assisting education through real-time learner analytics."

[15] O. Kiat Xin and D. Singh, "Development of learning analytics dashboard based on moodle learning management system," *IJACSA) International Journal of Advanced Computer Science and Applications*, vol. 12, p. 2021.

[16] S. Lee-Cultura, K. Sharma, and M. N. Giannakos, "Multimodal teacher dashboards: Challenges and opportunities of enhancing teacher insights through a case study," pp. 1–19, 01 2023.

[17] "Early warning system," 10 2022.

[18] "At-risk students early reporting system - edinsight," 09 2020.